

LONG TAIL SINGLE PIN COMPONENTS IN PCBA ASSEMBLY

Complete Guide to Types, Assembly Methods & Best Practices

Topics Covered:

- What are Long Tail Single Pin Components
- Types: Terminal Pins, Standoffs, Test Points, Connectors
- Assembly Methods: THT, SMT, Press-Fit, THR
 - Design & DFA Considerations
- Common Challenges & Solutions
- Inspection & Quality Standards

1. WHAT ARE LONG TAIL SINGLE PIN COMPONENTS?

Long tail single pin components are individual metal pins, posts, or terminals that extend significantly from the PCB surface (typically 6mm to 25mm+) and serve specific functions in PCBA assembly. Unlike multi-pin ICs or connectors, these are standalone components with a single electrical or mechanical connection point. They are widely used for:

- **Electrical Termination:** Power input, signal I/O, grounding points, battery contacts
- **Mechanical Support:** PCB standoffs, mounting posts, spacer pins
- **Test & Debug:** Test points, probe access points, programming headers
- **Interconnection:** Board-to-board stacking, daughterboard connectors, wire-wrap points
- **High-Current Paths:** Power distribution pins handling 10A to 100A+ loads

The '**long tail**' refers to the extended pin length protruding from the PCB, which enables connections to external systems, stacking of multiple boards, or attachment to enclosures and chassis. These components are critical in automotive, industrial, aerospace, and high-power electronics where robust mechanical and electrical connections are essential.

2. TYPES OF LONG TAIL SINGLE PIN COMPONENTS

Long tail single pin components come in various forms depending on their function, mounting method, and electrical requirements.

Type	Description	Typical Length	Assembly Method
Terminal Pins (Solder Tail)	Individual metal pins for wire termination or external connection. Often used for wire-to-board or wire-to-wire connections.	25mm	Hand Soldering, Wave Soldering, Reflow Soldering
PCB Standoffs (Male-Female)	Threaded or unthreaded posts that maintain spacing between PCBs and provide electrical isolation.	10-30mm	Press-Fit, Solder, Thermal Conduction
Test Point Pins	Small diameter pins (0.6–1.0mm) for probe access during testing or debugging.	12mm	(Solder), Silver Soldering
Press-Fit Pins	Solid pins pressed into plated through-holes without solder. Deform to create mechanical and electrical contact.	25mm	Press-Fit (No Solder)
Power Distribution Pins (Power Elements)	Heavy-duty pins for high-current applications. Can carry 10A to 50A+.	25mm	Press-Fit, TIG Welding
Pin Headers (Single Position)	Individual header pins for modular connections, jumpers, or programming.	15mm	Hot Air Soldering, THP
Guide / Alignment Pins	Non-electrical pins used purely for mechanical alignment in multi-board assemblies.	20mm	Press-Fit, THP (Solder)
BNC / RF Single Pin Connectors	Coaxial single-pin connectors for RF signal paths.	15mm	Hand Soldering, Reflow Soldering

Image Reference — Long Tail Pin Headers & Solder Tails:



0.64mm solder tail pin header connectors with extended long tails for through-hole PCB mounting

Image Reference — Single Pin PCB Mount Connectors:



BNC female single-pin right-angle PCB mount connectors with extended solder tails

3. ASSEMBLY METHODS FOR LONG TAIL SINGLE PIN COMPONENTS

Long tail single pin components can be assembled using multiple technologies depending on the component type, PCB design, and production volume.

3.1 Through-Hole Technology (THT)

The traditional method for long tail components. The pin is inserted through a drilled hole in the PCB and soldered on the opposite side.

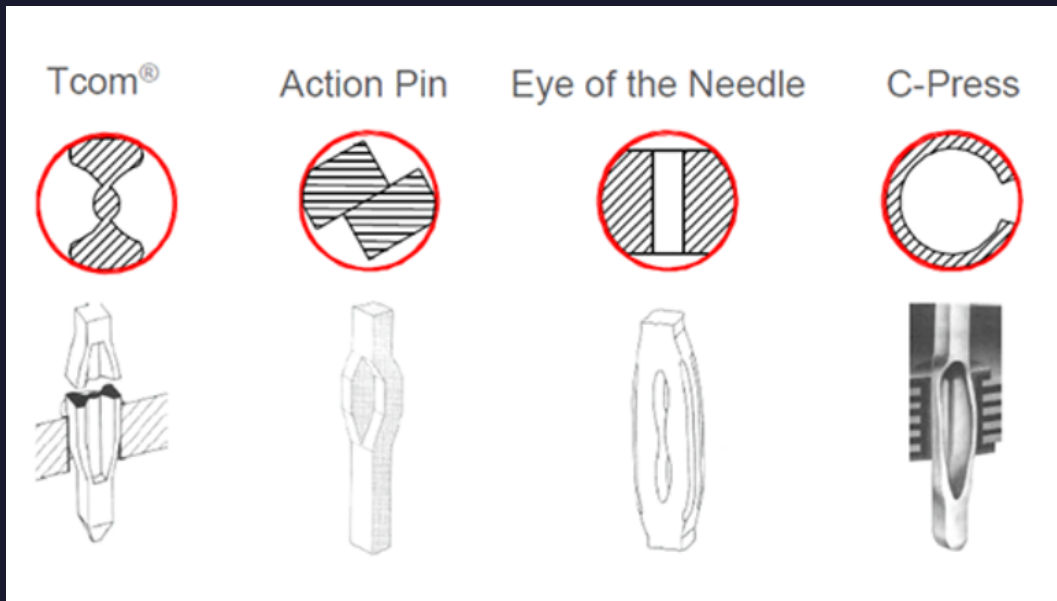
- **Wave Soldering:** PCB passes over a molten solder wave. Efficient for high-volume production but requires careful fixture design to protect SMT components on the same board.
- **Selective Soldering:** A solder nozzle targets specific through-hole joints individually. Ideal when SMT components are on the bottom side and wave soldering would damage them.
- **Hand Soldering:** Used for prototypes, low-volume runs, or large/heavy pins that automated systems cannot handle. Skilled technicians achieve joint strengths of 5–10 MPa.
- **Lead Trimming:** After soldering, leads are trimmed to 1–2mm protrusion per IPC-A-610 standards.

3.2 Press-Fit Technology

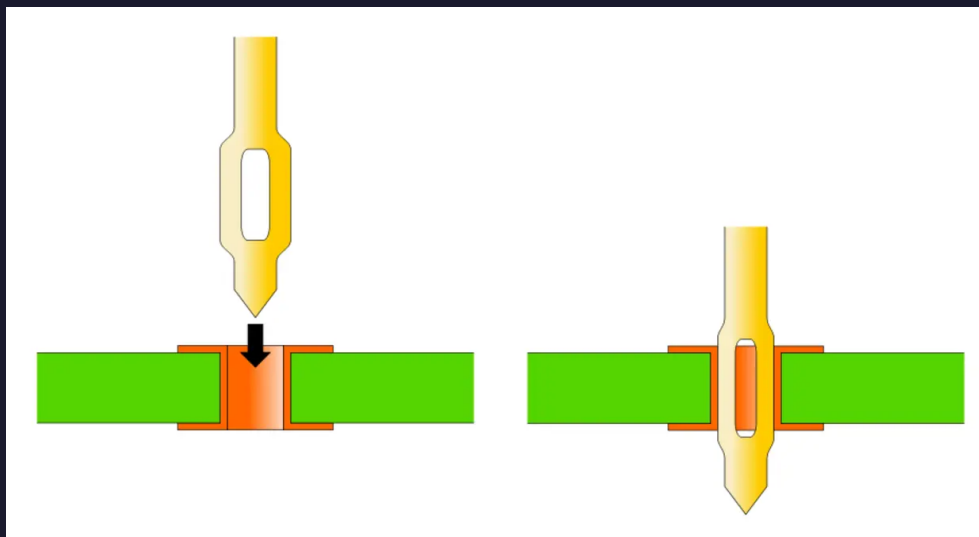
Press-fit pins are pressed into plated through-holes without any solder. The pin deforms (or the PCB hole deforms) to create a gas-tight, high-conductivity electrical connection with contact resistance as low as 0.1–0.2 mΩ.

- **Massive Press-Fit (MPFT):** Solid rectangular or square pins pressed into holes. The PCB hole deforms to match the pin shape. Best for high-current applications.
- **Flexible Press-Fit (FPFT):** The pin itself deforms (common shapes: 'needle eye', 'action pin', 'C-press'). Press-in force: 60–250N per pin. The pin springs back to create contact.
- **Advantages:** No solder = no thermal stress on PCB, no voids, excellent for high-vibration environments (automotive, aerospace), superior FIT rates.
- **Requirements:** PCB must be supported by a press-in pad during insertion. Hole plating must meet IPC specifications. Pin diagonal must exceed hole diameter for proper interference fit.

Image Reference — Press-Fit Pin Types:



Press-fit pin cross-sections: Tcom, Action Pin, Eye of the Needle, and C-Press designs



Press-fit technology: pin insertion before and after pressing into plated through-hole

3.3 Through-Hole Reflow (THR)

THR combines SMT and THT advantages. Solder paste is printed into through-holes using a stencil, the pin is placed, and the board passes through a standard reflow oven. This eliminates the need for a separate wave soldering step.

- **Process:** Solder paste fills the via → pin inserted → reflow soldered with other SMT components in one pass.
- **Benefits:** Single-pass assembly, compatible with standard SMT lines, no wave soldering required.
- **Challenges:** Heavy pins absorb significant heat during reflow — custom thermal profiles required. Solder paste volume must be carefully calculated to fill the hole completely.
- **Best For:** Mixed-technology boards where most components are SMT but a few through-hole pins are needed.

3.4 Surface Mount Technology (SMT) for Standoffs

Some standoffs and spacers are designed for SMT assembly. They have a flat base that sits on solder pads and is reflowed along with other SMDs.

- **SMT Standoffs:** Feature a snap-in or flat base design. Placed by pick-and-place machines and reflow soldered.
- **Advantages:** No drilled holes needed, can be placed anywhere on the PCB surface, automated assembly compatible.
- **Limitations:** Mechanical strength is lower than THT/press-fit. Bond interfaces are difficult to inspect visually. Mechanical failure can damage nearby components.
- **Best For:** Low-stress applications, consumer electronics, space-constrained designs.

4. DESIGN FOR ASSEMBLY (DFA) CONSIDERATIONS

4.1 Hole Size & Tolerance

Proper hole sizing is critical for reliable assembly of long tail single pin components:

Component Type	Lead Diameter	Recommended Hole	Plating
Standard Terminal Pin	0.6–0.8mm	0.9–1.0mm	25–35µm copper
Heavy Power Pin	1.0–1.6mm	1.3–1.8mm	35–50µm copper
Press-Fit Pin (MPFT)	1.0–2.0mm diagonal	0.9–1.6mm (smaller than pin)	25–40µm copper
Press-Fit Pin (FPFT)	0.8–1.5mm	0.8–1.4mm	20–35µm copper
PCB Standoff (THT)	M3–M4 thread	3.2–4.3mm	25–35µm copper
Test Point Pin	0.5–0.8mm	0.8–1.0mm	20–25µm copper

4.2 Spacing & Clearance Requirements

- **Component-to-Component:** Minimum 1.2–2mm clearance from nearby SMT components to prevent shadowing during wave soldering or interference during insertion.
- **Keep-Out Zones:** 4mm clearance around press-fit Powerelements to prevent damage to nearby components during insertion (60–250N force per pin).
- **Edge Distance:** Standoffs should be placed near board edges and corners for maximum stability. Avoid placing heavy pins near the board center where flex is highest.
- **Height Clearance:** Account for the full pin length plus any mating connector or wire bend radius in the enclosure design.

4.3 Thermal Considerations

Long tail pins, especially heavy power pins, act as heat sinks during soldering:

- **THR Soldering:** Heavy pins absorb heat from the reflow oven. Custom thermal profiles with extended soak zones may be needed to ensure complete hole fill.
- **Wave Soldering:** Large thermal mass pins may not reach proper soldering temperature. Pre-heating or dual-wave systems may be required.
- **Hand Soldering:** Use higher wattage irons (60–100W) with large chisel tips. Apply heat to both the pin and pad simultaneously for proper wetting.
- **Press-Fit:** No thermal stress — the major advantage for heat-sensitive boards or components.

4.4 Mechanical Stress & Vibration

Long tail pins create leverage points that can crack solder joints under vibration or thermal cycling:

- **Press-Fit is preferred** for high-vibration environments (automotive, aerospace) because the mechanical connection is independent of solder.
- **Solder Joint Reinforcement:** For THT pins in critical applications, add epoxy staking or conformal coating around the pin base.
- **Strain Relief:** Design wire bend radius and cable tie points so that external forces don't transfer to the pin solder joint.

- **Standoff Distribution:** Use multiple standoffs distributed uniformly to avoid stress concentration at any single point.

5. COMMON CHALLENGES & SOLUTIONS

Challenge 1: Pin Bending During Insertion

Long tail pins are prone to bending if not inserted perfectly straight. Bent pins cause misalignment, poor solder joints, and can damage the PCB hole plating.

Solutions:

- Use insertion guides or fixtures for high-volume production
- Specify pins with chamfered or bullet-shaped tips for easier insertion
- Implement pre-insertion inspection (visual or camera-based)
- For hand assembly, use pin straightening tools before soldering

Challenge 2: Incomplete Solder Fill in Through-Holes

Long tail pins with large thermal mass often result in cold solder joints or incomplete hole fill, especially in multilayer boards with thick copper plating.

Solutions:

- Increase pre-heat temperature and duration in wave/selective soldering
- Use THR process with adequate solder paste volume (stencil design critical)
- For hand soldering, apply heat to both sides of the board simultaneously
- Specify minimum 25µm hole plating for good solder wetting
- Consider press-fit as an alternative for high-reliability applications

Challenge 3: Co-Planarity Issues with SMT Standoffs

SMT standoffs may not sit flat on the PCB due to pad height variations or solder paste thickness inconsistencies, leading to tilted boards and poor mechanical stability.

Solutions:

- Use planar PCB finishes (ENIG, Immersion Silver) — avoid HASL for SMT standoffs
- Control solder paste thickness to 150µm ±25µm
- Design standoff pads 1mm larger than the component footprint
- For critical applications, use THT or press-fit standoffs instead

Challenge 4: Galvanic Corrosion with Dissimilar Metals

When copper-plated PCB holes mate with tin-plated or brass pins in press-fit applications, galvanic corrosion can occur in humid environments, increasing contact resistance over time.

Solutions:

- Use gold-plated or nickel-plated pins for critical applications
- Apply conformal coating over the press-fit area
- Specify compatible materials (e.g., CuZn39 brass pins with appropriate plating)
- Design for environmental sealing in outdoor or automotive applications

Challenge 5: Interference with SMT Components During Wave Soldering

When THT long tail pins and SMT components coexist on the same board, wave soldering the THT pins can expose SMT components to molten solder, causing shorts or damage.

Solutions:

- Place all THT components on one side and SMT on the opposite side
- Use selective soldering for mixed-technology boards
- Apply SMT adhesive (glue) to hold bottom-side SMT components during wave soldering
- Use THR process to eliminate wave soldering entirely

Challenge 6: Pin Length Variability

Long tail pins from different manufacturers may have length tolerances of $\pm 0.5\text{mm}$ or more, causing inconsistent protrusion heights in stacked assemblies.

Solutions:

- Specify tight length tolerances ($\pm 0.1\text{mm}$) in procurement
- Use pin headers with molded plastic bodies that act as built-in spacers
- Design adjustable mounting brackets for enclosure attachment
- Implement 100% pin length inspection before assembly

6. INSPECTION & QUALITY STANDARDS

6.1 Visual Inspection (AOI & Manual)

After assembly, long tail single pin components require specific inspection checks:

- **Solder Joint Quality:** Check for complete wetting, proper fillet formation, no voids or cold joints. IPC-A-610 Class 2/3 standards apply.
- **Pin Straightness:** Verify pins are perpendicular to the PCB surface ($\pm 2^\circ$ tolerance typical). Bent pins must be straightened or replaced.
- **Lead Protrusion:** Trimmed leads should be 1.0–2.0mm on the solder side per IPC-A-610.
- **Press-Fit Verification:** Check pin seating depth, no PCB cracking around holes, and correct insertion angle ($90^\circ \pm 3^\circ$).
- **Mechanical Stability:** Apply specified torque to standoffs and verify no rotation or loosening.

6.2 Electrical Testing

Long tail pins serving as electrical connections must pass:

- **Contact Resistance:** Press-fit connections: $<0.5\text{m}\Omega$. Soldered connections: $<1\text{m}\Omega$. Measure with 4-wire Kelvin method.
- **Insulation Resistance:** $>100\text{M}\Omega$ between adjacent pins and to ground.
- **High-Current Testing:** For power pins, apply rated current for 30 minutes and verify temperature rise $<40^\circ\text{C}$ above ambient.
- **ICT / Flying Probe:** Test point pins must be accessible and make reliable contact with test probes.

6.3 Mechanical Testing

For standoffs and structural pins:

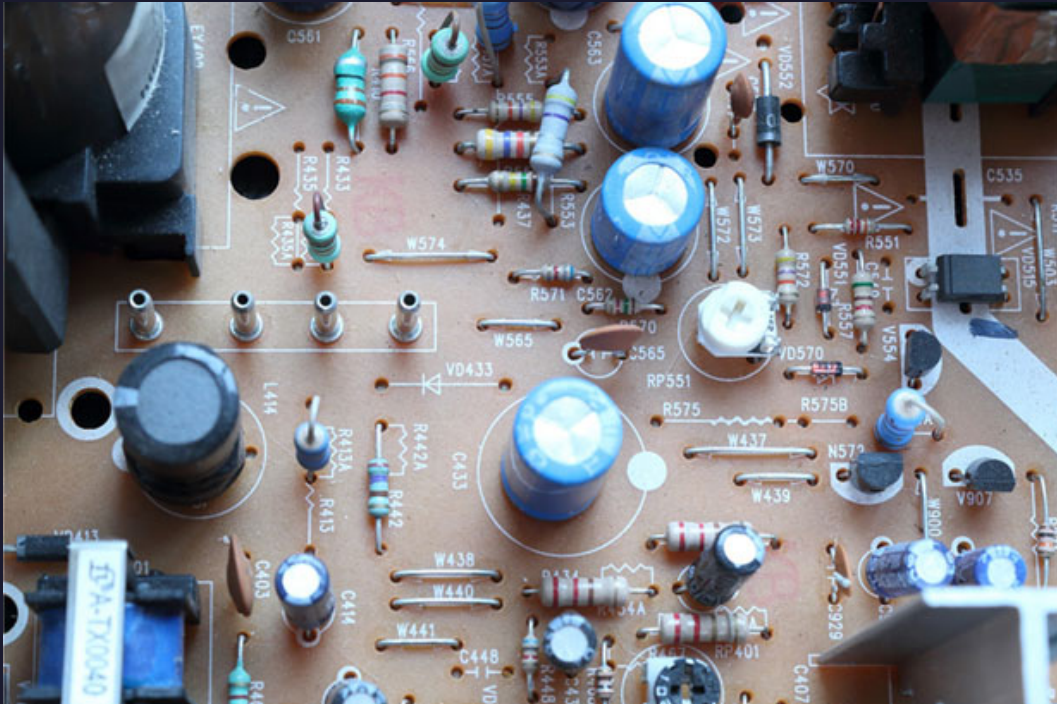
- **Pull-Out Force:** SMT standoffs: minimum pull-out force per manufacturer spec (typically 50–150N). THT/press-fit: significantly higher.
- **Torque Resistance:** Standoffs must withstand specified mounting torque without rotation or PCB damage.
- **Vibration Testing:** Per IEC 60068-2-6 or MIL-STD-810 for automotive/military applications.
- **Thermal Cycling:** -40°C to $+125^\circ\text{C}$, 1000 cycles minimum for automotive grade. Verify no solder joint cracking or press-fit loosening.

6.4 Relevant IPC Standards

Standard	Application
IPC-A-610	Acceptability of Electronic Assemblies — solder joint criteria for THT components
IPC-J-STD-001	Requirements for Soldered Electrical and Electronic Assemblies — soldering process requirements
IPC-7095	Design and Assembly Process Implementation for BGAs — includes THR process guidance
IPC-6012	Qualification and Performance Specification for Rigid PCBs — hole plating requirements
IPC-4761	Protection of Printed Board Structures — via filling for THR applications

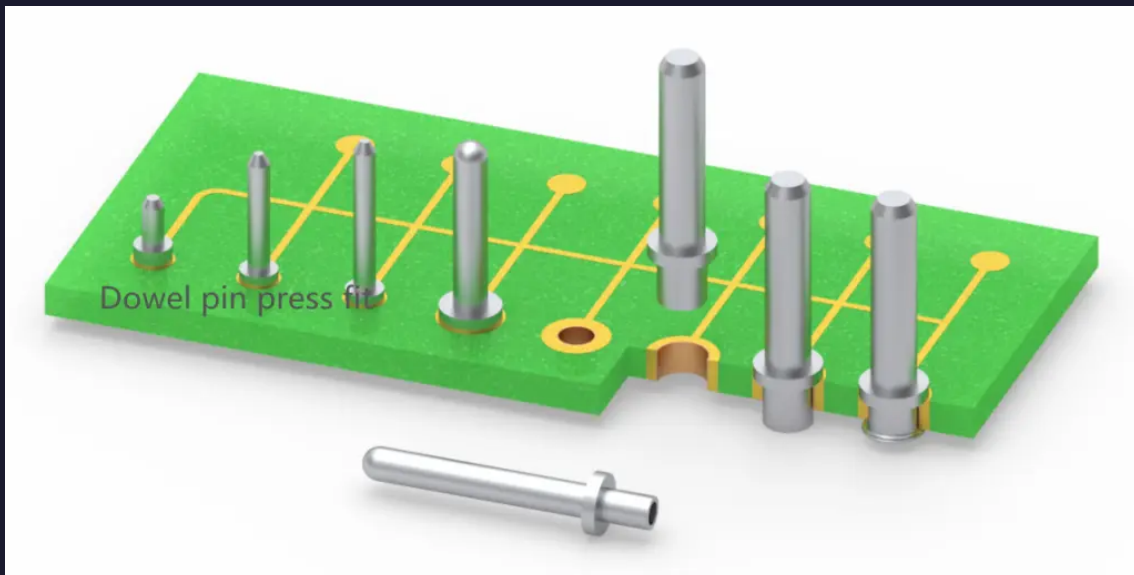
7. REFERENCE IMAGES

Through-Hole Assembly on PCB:



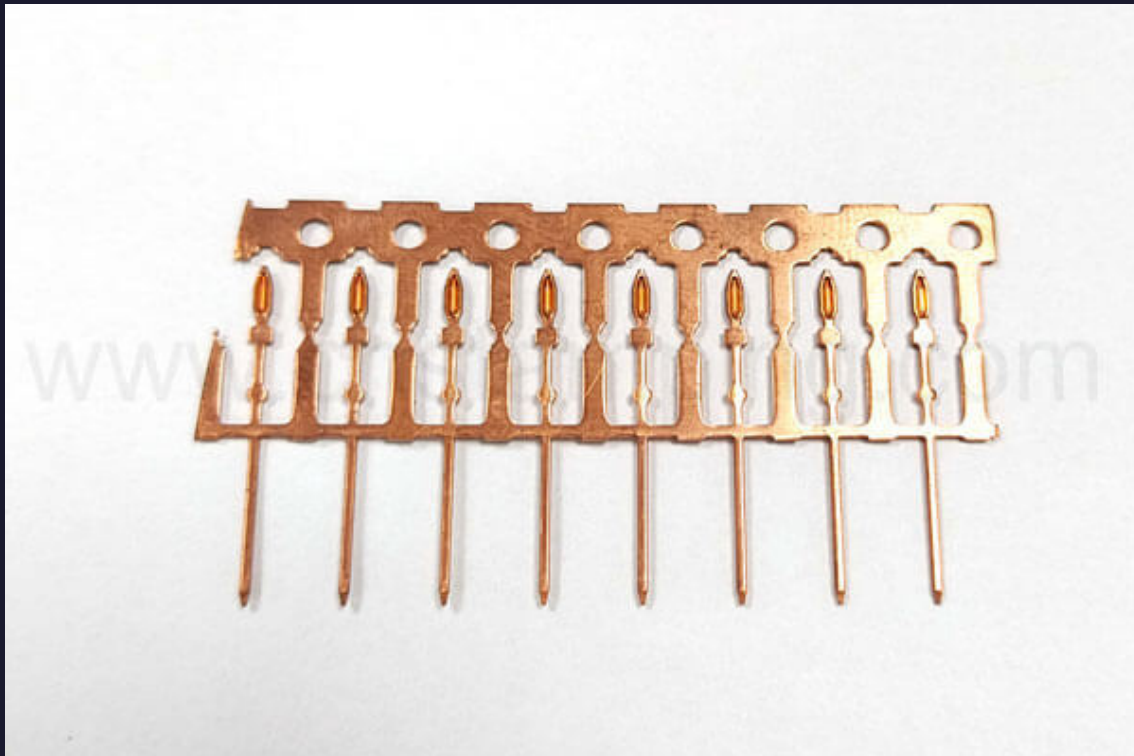
Through-hole components assembled on PCB including resistors, capacitors, and terminal pins

Press-Fit Dowel Pins on PCB:



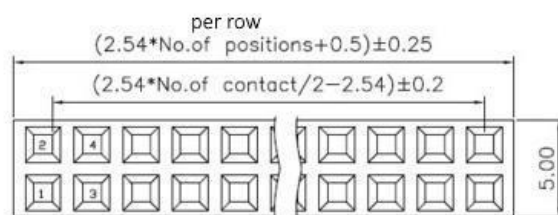
Dowel pin press-fit examples showing various pin sizes and PCB mounting configurations

Press-Fit Terminal Pins (Stamped):

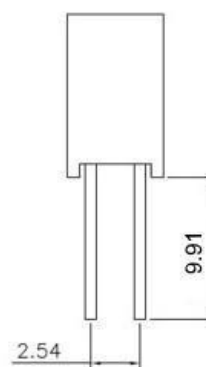
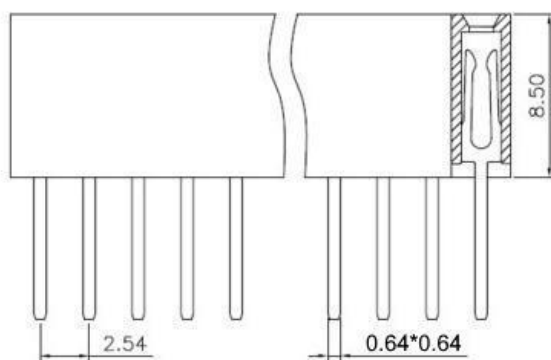


Stamped press-fit terminal pins on carrier strip for automated insertion

Long Tail Female Header with Dimensions:



Receptacles-Sockets PCB Mount
 .100" [2.54mm] Spacing Dual Row
 .390" [9.91mm] Tail Length
 .025" [0.64mm] Square Tails/Pins

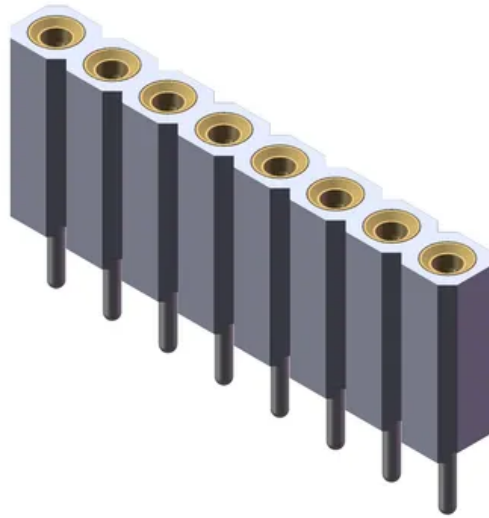


HWS11089

Long tail female header dimensional drawing showing 9.91mm tail length and 2.54mm pitch

Single Row PCB Mounting Socket Connector:

winconn®



Single row PCB mounting socket connector with extended solder tails for through-hole assembly

8. QUICK REFERENCE — ASSEMBLY METHOD SELECTION

Use this guide to select the appropriate assembly method for your long tail single pin component:

Application	Recommended Method	Key Considerations
High-volume production, all THT components	Wave Soldering	Design fixture to protect SMT components on bottom side
Mixed SMT + THT on same board	Selective Soldering or THR	Selective: targets specific joints. THR: single-pass reflow.
Prototype / low volume / repair	Hand Soldering	Use 60–100W iron, pre-heat heavy pins, trim to 1–2mm
High-vibration (automotive, aerospace)	Press-Fit (MPFT/FPFT)	No solder = no thermal stress. Gas-tight connection. 60–250N insert force.
High-current power distribution	Press-Fit or Heavy THT	MPFT for 50A+. Multiple parallel pins for current sharing.
Board-to-board stacking	THT Pin Headers + Standoffs	Use guide pins for alignment. Account for total pin length.
Space-constrained, no holes allowed	SMT Standoffs	Planar finish required. Lower mechanical strength than THT.
Test points, programming headers	THT (Solder) or SMT	Ensure probe accessibility. Keep away from tall components.

Key Takeaways:

1. Long tail single pin components are essential for power, mechanical support, and interconnection in PCBA.
2. Press-fit technology offers the highest reliability for high-vibration and high-current applications without soldering.
3. THR (Through-Hole Reflow) is the best choice for mixed-technology boards, eliminating the need for wave soldering.
4. Always account for thermal mass when soldering heavy pins — custom profiles and higher wattage tools are often needed.
5. Design for proper hole size, keep-out zones, and mechanical stress relief to ensure long-term reliability.